

Anonymity in Networks

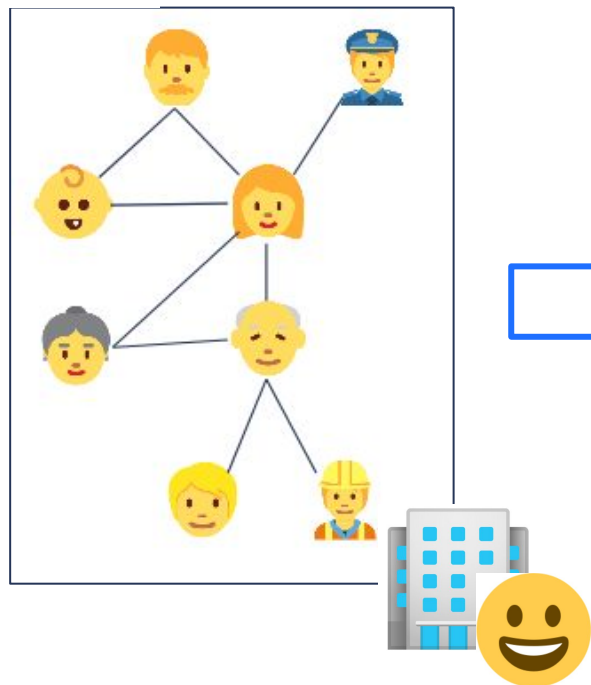
Rachel de Jong

Supervisors: Mark van der Loo, Frank Takes

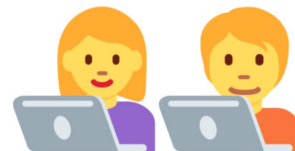
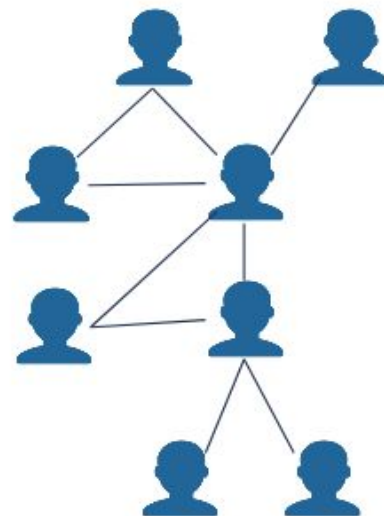
NetSciNL symposium | 18-10-2024



Sharing networks and privacy

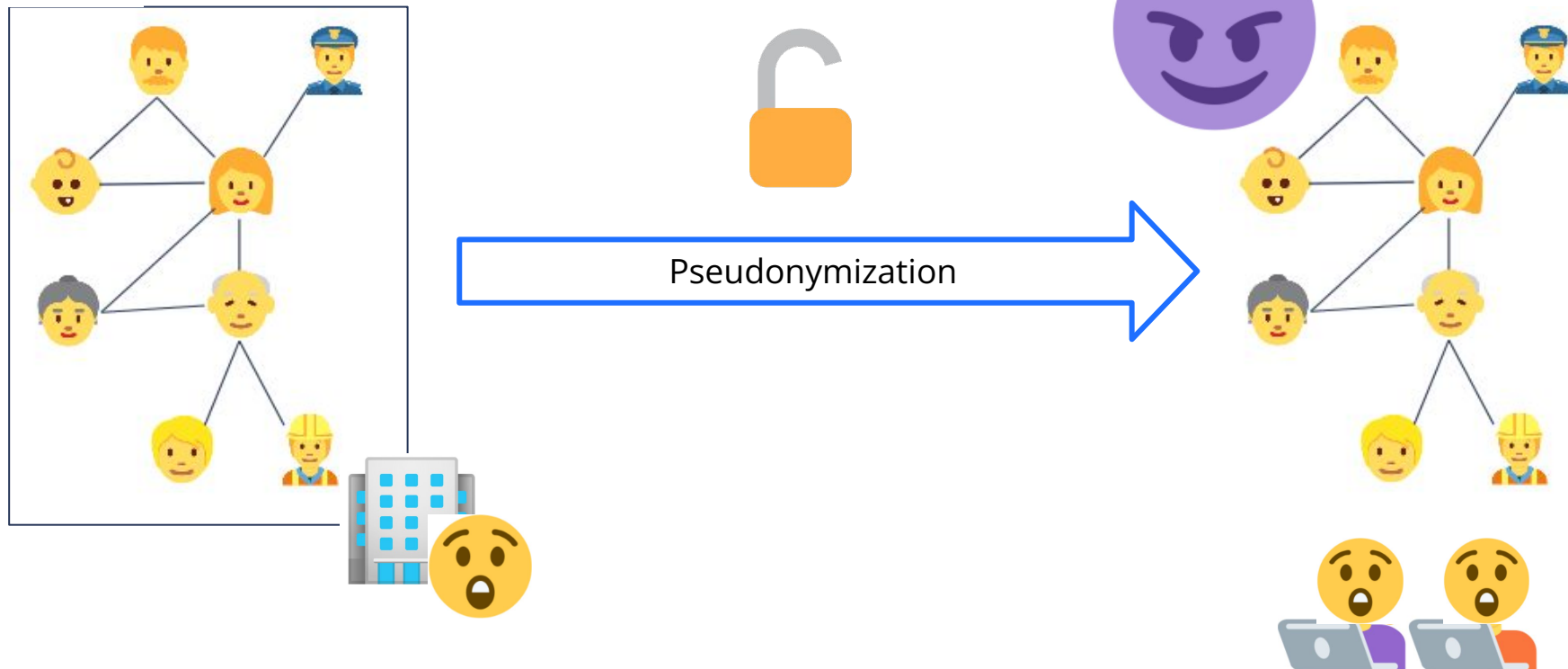


Pseudonymization



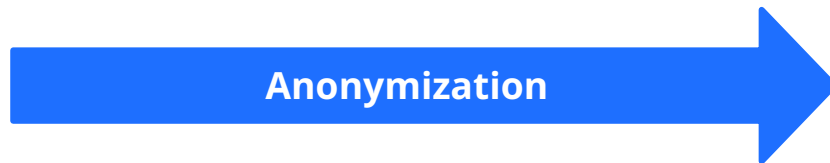
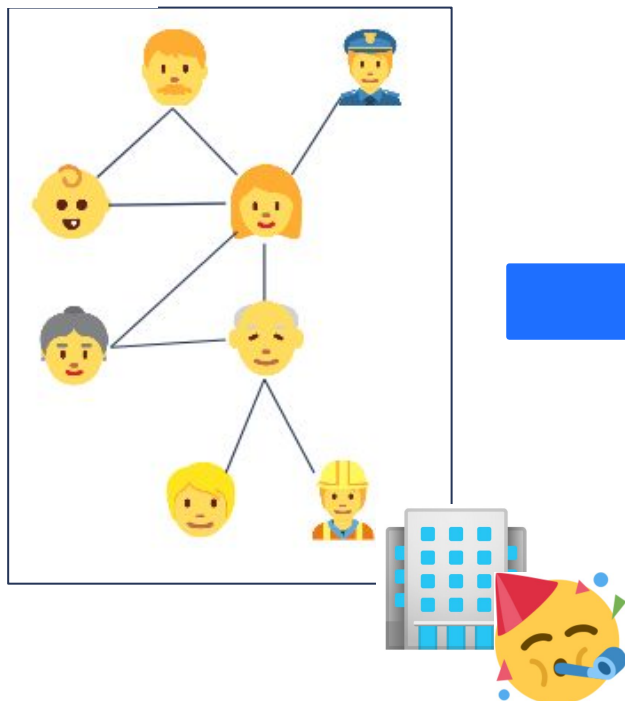


Sharing networks and privacy

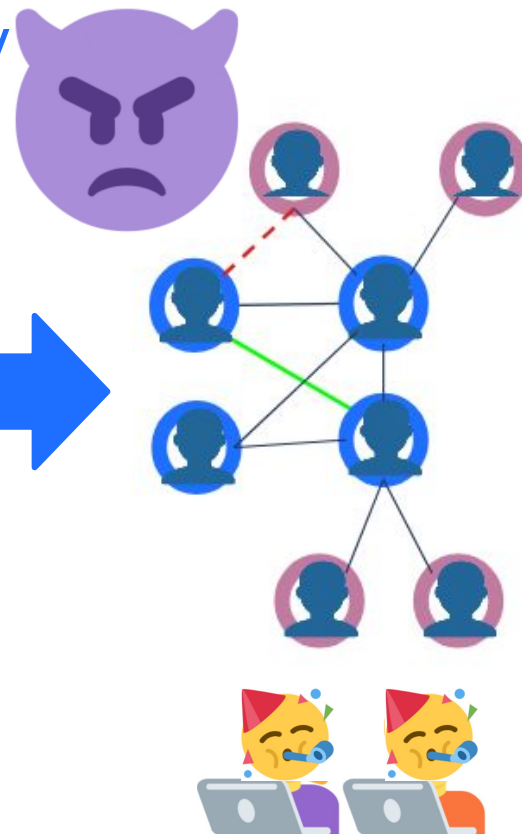




Sharing networks and privacy



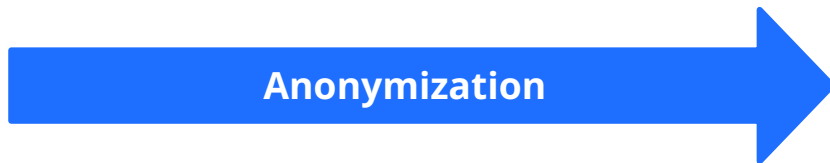
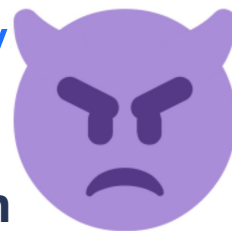
Deleting edges - - - - -
Adding edges ————



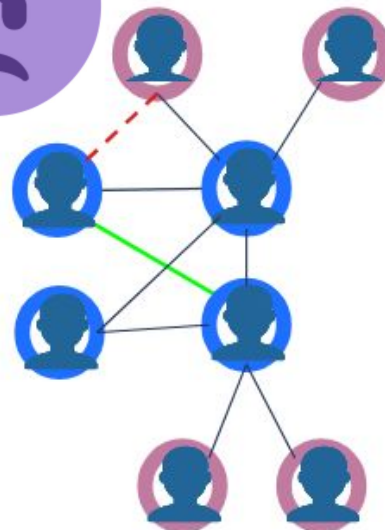
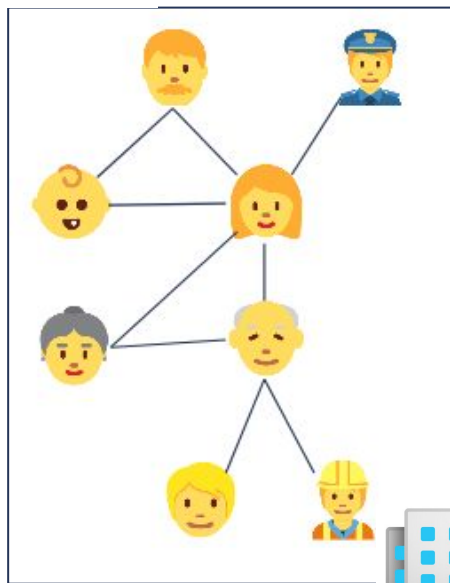


Sharing networks and privacy

1. **Anonymity measures**
2. **Anonymization algorithm**
3. Utility?



Deleting edges ---
Adding edges ---



Not unique

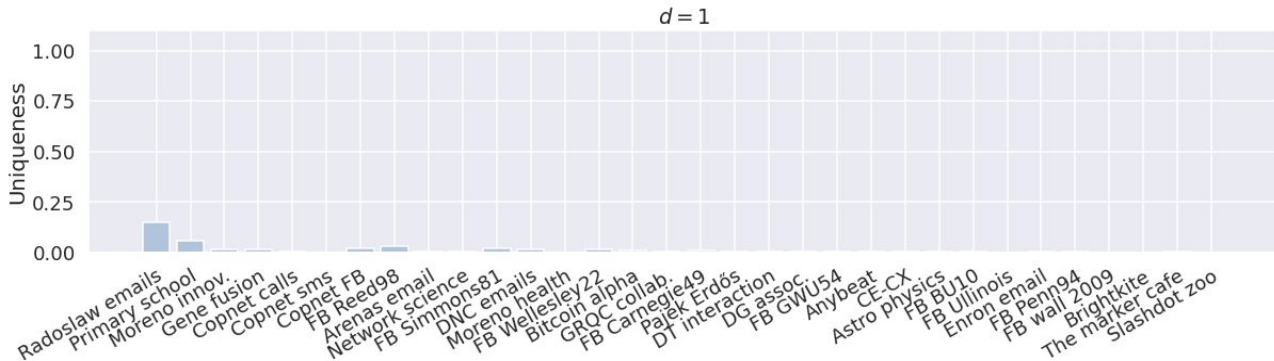


Anonymous*

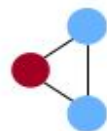
*For now



Anonymity measures



$|V|: \sim 100 - 3M$ $|E|: \sim 100 - 18M$

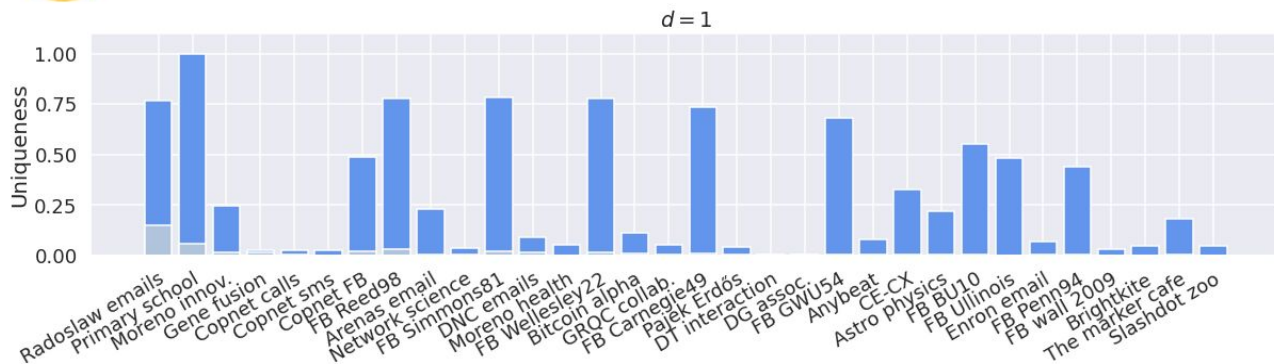


Degree 2

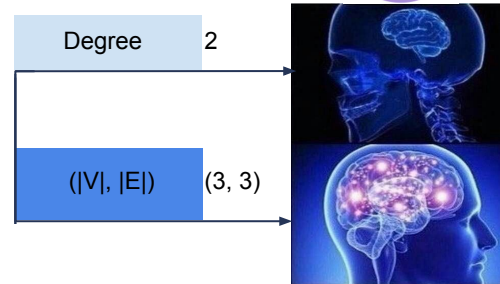
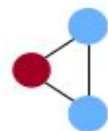




Anonymity measures

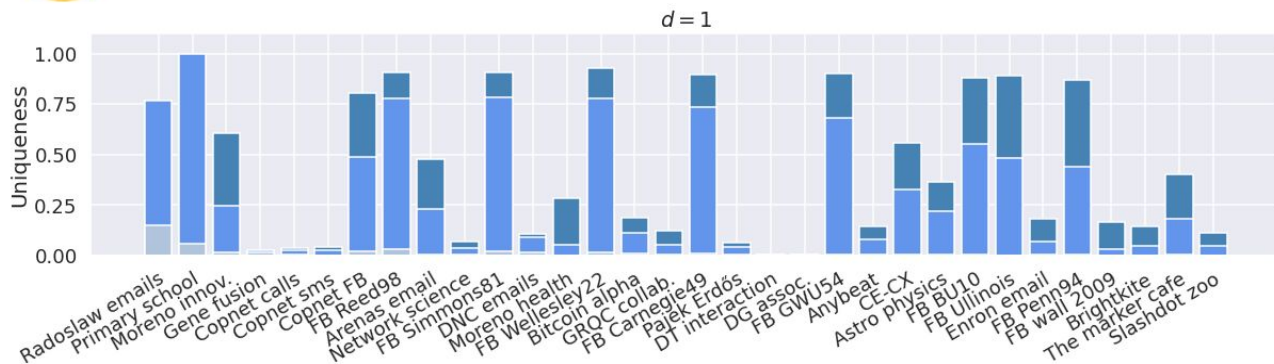
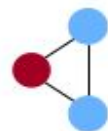


$|V|$: ~100 - 3M $|E|$: ~100 - 18M

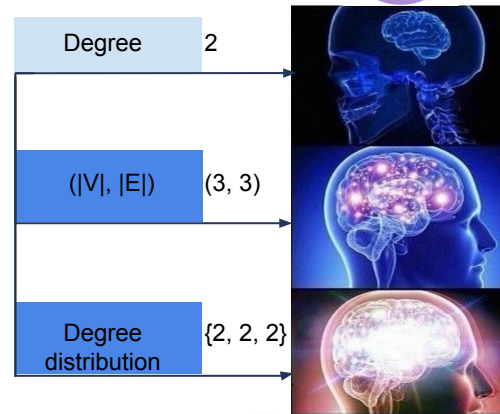




Anonymity measures

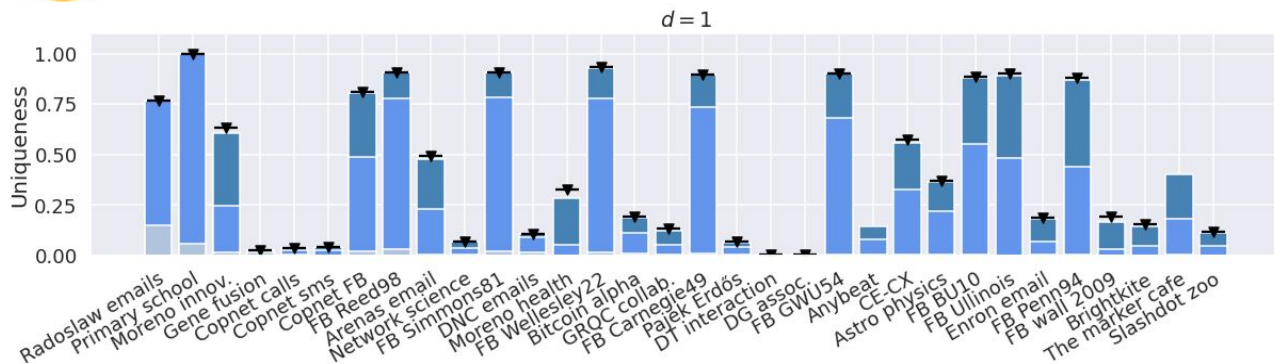
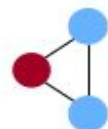


$|V|$: ~100 - 3M $|E|$: ~100 - 18M

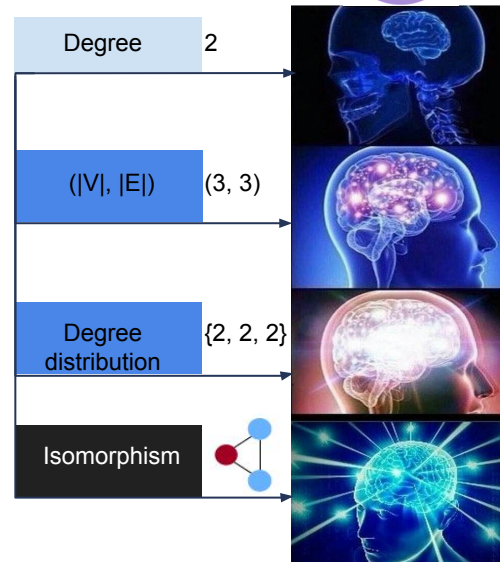




Anonymity measures

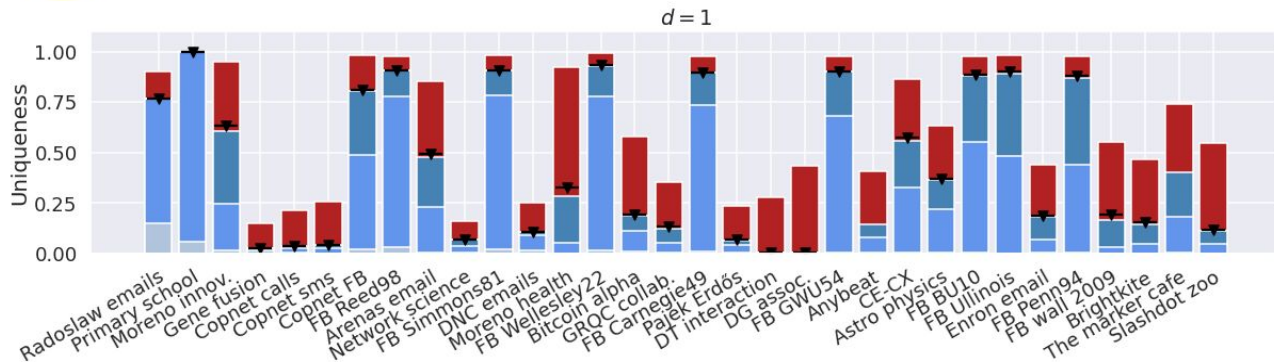


$|V|$: ~100 - 3M $|E|$: ~100 - 18M



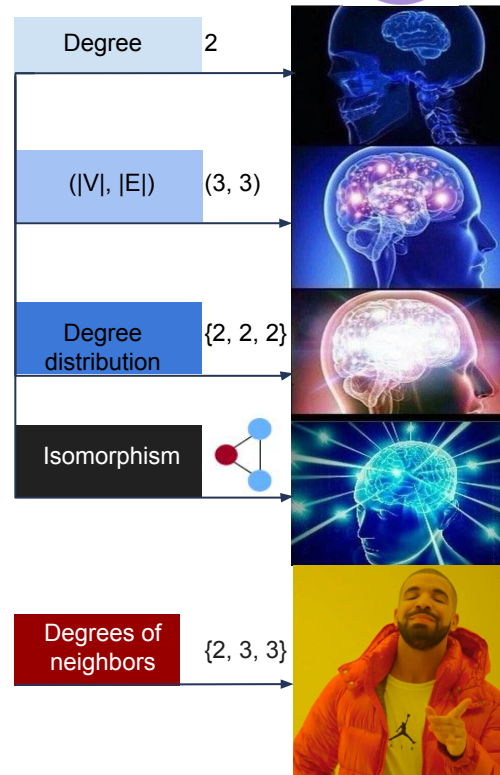
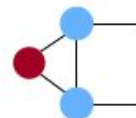
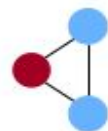


Anonymity measures



$|V|$: ~100 - 3M $|E|$: ~100 - 18M

Imprecise information but looking further
is more effective!



Anonymization!

Deleting few edges



High utility*

*5% for now, better measures for utility would be interesting for future work :)



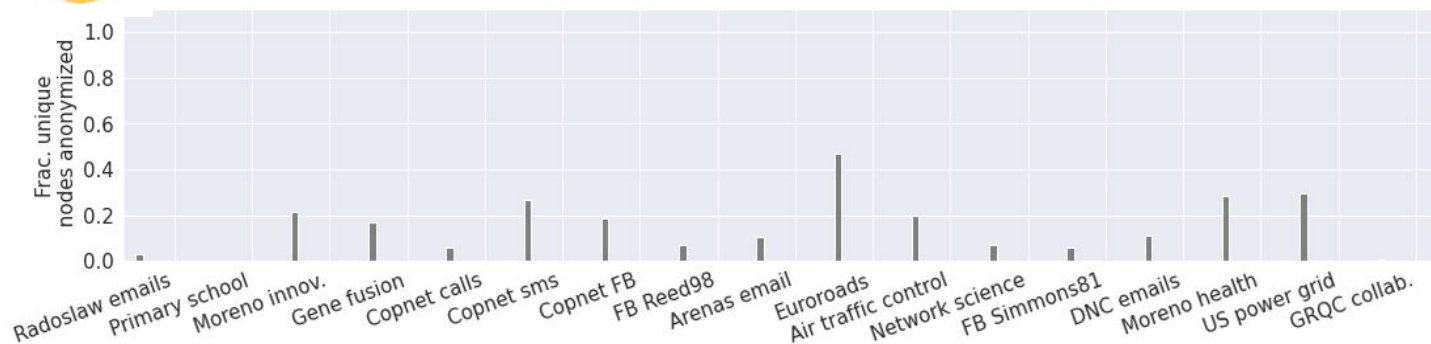
Anonymization



Deleting 5% of edges



Random



$|V|$: ~100 - 5.000 $|E|$: ~300 - 33.000

For more details
see our paper!





Anonymization

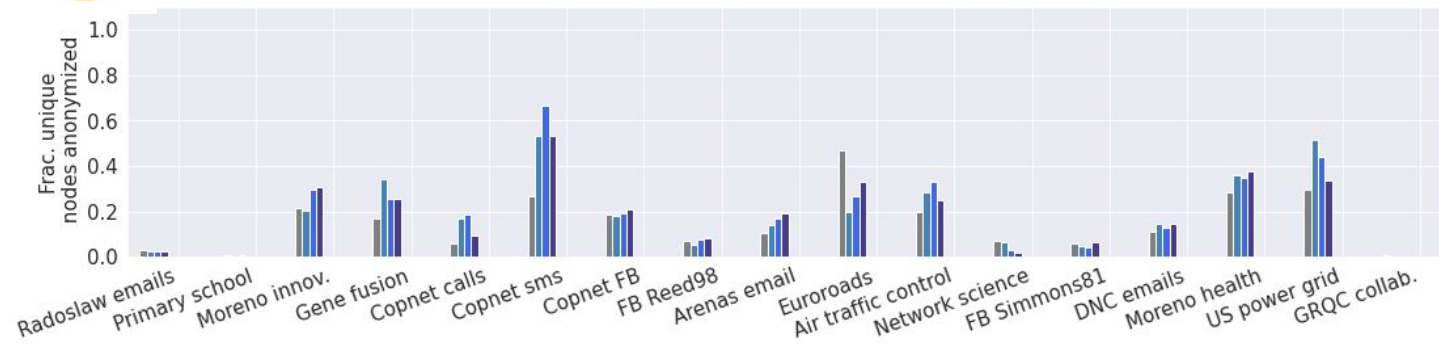


Deleting 5% of edges



Random

Structure
based



$|V|$: ~100 - 5.000 $|E|$: ~300 - 33.000

For more details
see our paper!

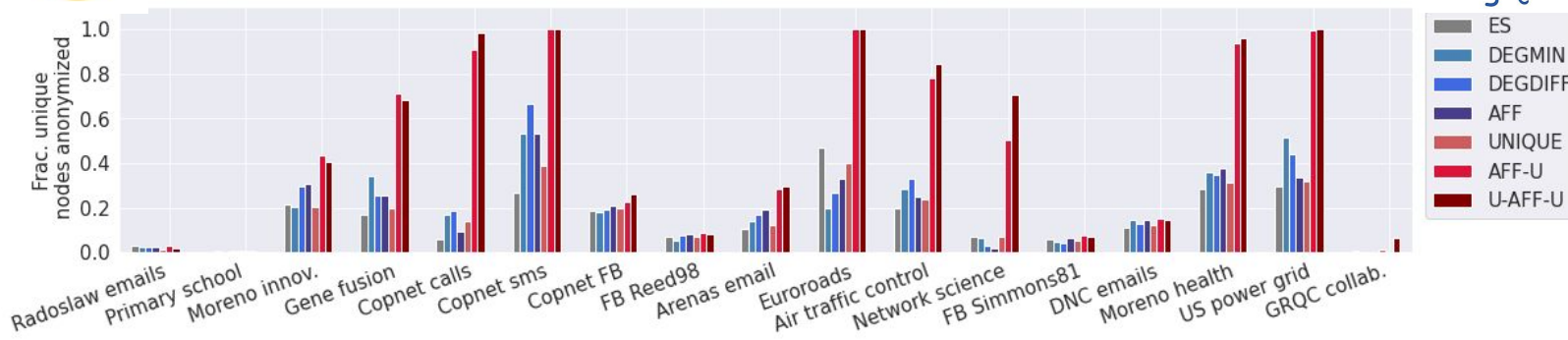




Anonymization



Deleting 5% of edges



Random

Structure based

Uniqueness based



$|V|$: ~100 - 5.000 $|E|$: ~300 - 33.000

For more details
see our paper!





Conclusion

- Anonymization → share data privacy aware
- Measure = attacker scenario
 - **Imprecise information but looking further** is more effective
- Anonymization:
 - Account for **which nodes are unique!**

Much more to do!

- Design better **anonymization algorithms**
- How to account for **utility?**

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ANONET package: implement and test your own anonymization algorithms!

github.com/RacheldeJong

